COURSE SYLLABUS

Python Programming

24 modules | 96 sessions | live classes, recordings, assignments and placement support

01. Orientation and Python setup (Week 1)

- Installing Python, VS Code and the terminal
- Running scripts, the REPL and notebooks
- Virtual environments and pip
- How the course and assignments work

02. Python syntax essentials (Week 1)

- Variables, numbers, strings and booleans
- Operators, input and output
- Conditionals and loops
- Writing clean, readable code from day one

03. Collections in depth (Week 2)

- Lists, tuples and slicing
- Dictionaries and nested data
- Sets and when they help
- List and dict comprehensions

04. Functions and modules (Week 2)

- Parameters, defaults and keyword arguments
- Return values, scope and closures
- Importing modules and packages
- Writing your first reusable utility module

05. Strings and text processing (Week 3)

- Formatting, f-strings and slicing
- Regular expressions for validation
- Cleaning messy real world text
- Lab: parsing a CSV report

06. Files, JSON and CSV (Week 3)

- Reading and writing text files safely
- Working with JSON APIs and config
- CSV files and the csv module
- Error handling with try, except and finally

07. Object oriented Python (Week 4)

- Classes, objects and attributes
- Methods, properties and dunder methods
- Inheritance and composition
- Dataclasses and type hints

08. Exceptions and logging (Week 4)

- Built in exception types
- Custom exceptions and error contracts
- Logging levels and handlers
- Debugging with breakpoints

09. Testing with pytest (Week 5)

- Writing your first test
- Fixtures and parametrised tests

- Mocking external services
- Coverage and test organisation

10. Standard library toolkit (Week 5)

- datetime, pathlib and os
- collections and itertools
- argparse for command line tools
- Building a CLI tool end to end

11. Working with APIs (Week 6)

- HTTP basics and the requests library
- Authentication, headers and pagination
- Retries, timeouts and rate limits
- Lab: build a data pulling script

12. Web scraping responsibly (Week 6)

- HTML parsing with BeautifulSoup
- Selectors and pagination
- Robots.txt, terms and politeness
- Saving scraped data to CSV and SQLite

13. SQL and SQLite (Week 7)

- Tables, keys and relationships
- SELECT, JOIN and aggregate queries
- Using SQLite from Python
- Parameterised queries and safety

14. Pandas fundamentals (Week 7)

- Series, DataFrames and indexing
- Filtering, sorting and grouping
- Merging tables and handling missing data
- Lab: sales report from raw CSV

15. Data visualisation (Week 8)

- Matplotlib charts that read well
- Seaborn for statistical plots
- Plotly interactive dashboards
- Choosing the right chart

16. NumPy essentials (Week 8)

- Arrays, shapes and broadcasting
- Vectorised operations and performance
- Random numbers and simulation
- Moving between NumPy and Pandas

17. Building web apps with Flask (Week 9)

- Routes, templates and static files
- Forms, validation and flash messages
- Blueprints and app structure
- Deploying a Flask app

18. Django fundamentals (Week 10)

- Project and app structure
- Models, migrations and the admin
- Views, templates and URL routing
- Forms and authentication

19. Django REST APIs (Week 10)

- Serializers and viewsets
- Permissions and throttling
- Pagination, filtering and search
- Documenting the API

20. Automation and scheduling (Week 11)

- Email, PDF and Excel automation
- Watching folders and reacting to files
- Cron and task schedulers
- Building an internal reporting bot

21. AI with Python (Week 11)

- Calling language model APIs
- Prompt design and structured output
- Embeddings and simple semantic search
- Cost control and guardrails

22. Project 1: data report automation (Week 12)

- Requirement and data source mapping
- Cleaning pipeline and validation
- Charts, summary and PDF export
- Scheduling and delivery

23. Project 2: REST API with auth (Week 12)

- Data modelling and migrations
- Auth, roles and permissions
- Tests and API documentation
- Deployment and monitoring

24. Interview preparation (Week 13)

- Python interview question bank
- Coding round practice
- Explaining projects and trade offs
- Mock interview and feedback

Fees, batch dates and mode (online / classroom): confirm the current numbers on the website or with the admissions team before enrolling. Yantrix Technology, Pune.